Enforcement Reports API Usage Documentation

Welcome to the Enforcement Reports API Usage Documentation page. The documentation provides information regarding how the API retrieves recall information using the various REST Endpoints for the Enforcement Reports application, called iRES (internet Recall Enterprise System). REST endpoints provide a 'RESTful' web service that generates an action to retrieve or act upon a specific piece of information that resides in a database.

To use the API REST Endpoints, enter the Authorization-User and Authorization-Key by clicking the iRES Authorize Button at the top. Please submit authorization key requests using the online OII Unified Logon application.

Each REST Endpoint is documented below and allows users that have entered the Authorization-User and Authorization-Key on this page to see how the request URL is constructed for each REST Endpoint and what each returns. For information on the definition of each result column, please visit the API definitions. When developing a method to utilize this API not using this page, below are some points that may help explain why results are not being returned as expected:

- The API cannot be used with a typical web browser (e.g. cutting and pasting the request URL into the URL bar of the browser). This is because the API request requires the use of the Authorization-User and Authorization-Key headers to be transmitted with the request.
- API calls are cached, so when constructing the string for an API call unique signatures should be used (as shown in the request URL examples below using "signature="). For example, if you send a request which results in a 401 message (i.e. you put the wrong Authorization-User and Authorization-Key in) and you try that same request with the same signature it will still return a 401 message even with the right Authorization-User and Authorization-Key information entered. It is recommended that signatures should use a date and time numeric component to make the request unique.
- The API server has a specified Access-Control-Allow-Origin and so API calls using some methods may not work. For example, regular JavaScript/JQuery may not work due to web browser security designed to not allow cross-domain calls. This is not an issue when calling the API using server-side code such as Java, PHP, or ColdFusion that make server-level HTTP requests.
- It is recommended that API calls use TLS 1.2 to avoid any connection related errors.
- For POST endpoints the header Content-Type should have the value application/x-www-form-urlencoded.
- The following are code examples for using the POST operation in various languages:

  Python

  PHP

  Java

  C#

For requests for assistance regarding the iRES API, please e-mail OIIWebApplicationTechTeam@fda.hhs.gov.

Following is an example of the iRES API REST Endpoint. You will need to click on the REST Endpoint title to expand the window.

GET /search/producttypes

Upon clicking, the REST Endpoints display the six sections including Implementation Notes, Response Class, Parameters, Response Messages, Request URL, and Response Body.

Implementation Notes - Gives a brief description of what data is accessed and displayed.

Response Class - Indicates the response class description of the API.

Parameters - Allows entering query parameters for the endpoint in the text box.

Response Messages - Is a legend that shows the API return codes and their meaning.

Request URL - Shows the URL of the requested API path.

Response Body - Shows the response of the API request.

Finally, notice the **Try it out!** button. Once you enter a specific value for the selected REST Endpoint in the Parameters field, click the **Try it out!** button. The results are displayed in the Response Body window.

REST Endpoints

GET /search/producttypes

Implementation Notes

Gets the list of all the available Product Types within the iRES application in JSON format including the following columns: centercd, producttypeshorttxt.

Response Class

Model | Example Value

```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```

Response Content Type application/json

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request

GET /recalls/event/{eventid}

Implementation Notes

Gets the event information for a given event ID in JSON format including the following columns: recalleventid, firmcitynam, firmcountrynam, firmline1adr, firmline2adr, firmstatecd, firmpostalcd, phasetxt, recallinitiationdt, firmlegalnam, voluntarytypetxt, distributionareasummarytxt, centercd, createdt, firmstateprvncnam, centerclassificationdt, terminationdt, initialfirmnotificationtxt, distributionpatternindicator, distributionpatternshort, determinationdt, enforcementreportdt, eventlmd, firmfeinum, firmsurvivingnam, firmsurvivingfei.

Response Class

Model | Example Value

```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
```

```
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```


Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
eventid	(required)	**ID of event**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
421	Event Id provided should be numeric

GET /recalls/product/{productid}

Implementation Notes

Gets the product information for a given product ID in JSON format including the following columns: productid, productdescriptiontxt, recalleventid, productshortreasontxt, producttypeshort, recallnum, productdistributedquantity, centerclassificationtypetxt, productdescriptionindicator, productdescriptionshort, recallreasonindicator, recallreasonshort, codeinfoindicator, enforcementreportdt, productlmd, determinationdt, codeinformation, codeinfoshort.

Response Class

Model | Example Value


```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```


Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
productid	(required)	**ID of product**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
427	Product Id provided should be numeric

GET /recalls/eventproducts/{eventid}

Implementation Notes

Gets all the Products for a given event ID in JSON format including the following columns: productid, productdescriptiontxt, recalleventid, productshortreasontxt, producttypeshort, recallnum, productdistributedquantity, centerclassificationtypetxt, productdescriptionindicator, productdescriptionshort, recallreasonindicator, recallreasonshort, codeinfoindicator, enforcementreportdt, determinationdt, firmfeinum, firmsurvivingnam, firmsurvivingfei, firmlegalnam, codeinformation, codeinfoshort.

Response Class

Model | Example Value

```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```

Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
eventid	**(required)**	**ID of event**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
421	Event Id provided should be numeric

GET /search/codeinfo/{productid}

Implementation Notes

Gets the code information for a given product ID in JSON format including the following columns: productid, codeinformation.

Response Class

Model | Example Value

```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```


Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
productid	(required)	**ID of product**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
427	Product Id provided should be numeric

GET /search/producthistory/{productid}

Implementation Notes

Gets the product history for a given product ID in JSON format including the following columns: eventid, productid, fieldname, productlmd, newvalue, oldvalue.

Response Class

Model | Example Value


```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```


Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
productid	(required)	**ID of product**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
427	Product Id provided should be numeric

GET /search/eventproducthistory/{eventid}

Implementation Notes

Gets the event history for a given event ID in JSON format including the following columns: eventid, productid, fieldname, eventlmd, newvalue, oldvalue.

Response Class

Model | Example Value

```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```

Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
eventid	(required)	**ID of event**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
421	Event Id provided should be numeric

GET /search/pressreleaseurls/{eventid}

Implementation Notes

Gets all the press releases for a given event ID in JSON format including the following columns: recalleventid, pressreleasetype, pressreleaseissuedt, pressreleaseurl.

Response Class

Model | Example Value


```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```


Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
eventid	(required)	**ID of event**	path	integer

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
421	Event Id provided should be numeric

POST /recalls/

Implementation Notes

Gets all the recalls based on the criteria using a payload.

Payload Structure - The structure can be in any order

{"displaycolumns":"","filter":"[]","start":1, "rows":10, "sort":"","sortorder":""}

The "payload" parameter is added in the request body with the value being in the payload structure noted above and with examples below. The payload should be added as form data or json.

Display columns - A list of columns (without spaces) that are retrieved from the dataset. Possible values:

productid, recalleventid, producttypeshort, firmcitynam, firmcountrynam, firmline1adr, firmline2adr, firmstatecd, firmpostalcd, phasetxt, recallinitiationdt, firmlegalnam, voluntarytypetxt, distributionareasummarytxt, centercd, firmstateprvncnam, centerclassificationdt, terminationdt, initialfirmnotificationtxt, centerclassificationtypetxt, enforcementreportdt, firmfeinum, firmsurvivingnam, firmsurvivingfei, eventlmd, productdescriptiontxt, productshortreasontxt, recallnum, productdistributedquantity, determinationdt, postedinternetdt, rid, codeinformation

Start - A numeric value used to indicate the beginning of a dataset, used as an offset for pagination of smaller datasets. This is a required parameter.

Rows - A numeric value used to indicate the number of rows requested from the dataset. This is a required parameter.

The payload retrieves 5000 records at a time if the "Code Information" column is not specified in the "displaycolumns" parameter.

The payload retrieves 2500 records at a time if the "Code Information" column is specified in the "displaycolumns" parameter.

Sort - A string containing the field that sorting is primarily applied on the dataset. Sort can accept only one value. This is a required

parameter. Column names used in the sort are case insensitive. The values that can be used to sort are:

productid, recalleventid, producttypeshort, firmcitynam, firmcountrynam, firmline1adr, firmline2adr, firmstatecd, firmpostalcd, phasetxt, recallinitiationdt, firmlegalnam, voluntarytypetxt, distributionareasummarytxt, centercd, firmstateprvncnam, centerclassificationdt, terminationdt, initialfirmnotificationtxt, centerclassificationtypetxt, firmfeinum, firmsurvivingnam, firmsurvivingfei, eventlmd, productdescriptiontxt, productshortreasontxt, recallnum, productdistributedquantity, determinationdt, enforcementreportdt, postedinternetdt

SortOrder - A string that specifies the order in which sorting is applied on the dataset. SortOrder can accept only one value. This is a required parameter. Possible values are either "asc" for sorting by ascending order or "desc" for sorting by descending order.

Filter - An array of filters in key value pairs using single quotes that can be applied over the dataset. Filters are optional but a blank filter parameter is required for the payload. Filters can accept only one value at a time. Column names used as filters are case insensitive and values input will perform a full text search. The following are filters that can accept multiple values and the values they can accept:

● 'phasetxt':['terminated','completed','ongoing']

● 'centercd':['CBER','CDER','CDRH','CFSAN','CTP','CVM','HFP','OCS']

● 'centerclassificationtypetxt':['3','2','1','NC']

The values that can be used to filter are:

productid, recalleventid, producttypeshort, firmcitynam, firmcountrynam, firmline1adr, firmline2adr, firmstatecd, firmpostalcd, phasetxt, recallinitiationdtfrom, recallinitiationdtto, firmlegalnam, voluntarytypetxt, distributionareasummarytxt, centercd, firmstateprvncnam, centerclassificationdtfrom, centerclassificationdtto, terminationdtfrom, terminationdtto, initialfirmnotificationtxt, centerclassificationtypetxt, firmfeinum, firmsurvivingnam, firmsurvivingfei, eventlmdfrom, eventlmdto, productdescriptiontxt, productshortreasontxt, recallnum, productdistributedquantity, determinationdtfrom, determinationdtto, enforcementreportdtfrom, enforcementreportdtto, postedinternetdtfrom, postedinternetdtto

Examples of Payloads to retrieve Recalls:

1. Payload to retrieve all Product IDs

{"displaycolumns":"", "filter":"[]", "start":1, "rows":5000, "sort":"productid","sortorder":"asc"}

2. Payload to retrieve columns stored as Text BLOBS (large text fields) in the output

{"displaycolumns":"productdescriptiontxt,productshortreasontxt,productdistributedquantity,codeinformation","filter":"[]", "start":1, "rows":2500, "sort":"productid", "sortorder":"asc"}

3. Payload to retrieve Recalls where Product Description, Reason for Recall, Classification, or Code Information has been updated

{"displaycolumns":"eventlmd,recalleventid,firmcitynam","filter":"[{'eventlmdfrom':'07/24/2018'} , {'eventlmdto':'09/24/2018'}]","start":1,"rows":50,"sort":"productid","sortorder":"asc"}

4. Payload to filter multiple values

{"displaycolumns":"productdescriptiontxt,productshortreasontxt,productdistributedquantity,codeinformation,phasetxt","filter":"[{'phasetxt': ['terminated','completed','ongoing']}]", "start": 1,"rows": 2500, "sort": "productid", "sortorder": "asc"}

{"displaycolumns":"recalleventid,centerclassificationtypetxt,firmcitynam", "filter":"[{'centerclassificationtypetxt':['3','2','1','NC']}]", "start":1,"rows":75,"sort":"productid","sortorder":"asc"}

{"displaycolumns":"recalleventid,centerclassificationtypetxt,firmcitynam,centercd","filter":"[{'centercd':['CBER','CFSAN']}]", "start":1,"rows":75,"sort":"productid","sortorder":"asc"}

5. Payload to filter by a single value

{"displaycolumns":"productdescriptiontxt,productshortreasontxt,productdistributedquantity,codeinformation","filter":" [{'productdescriptiontxt':'Colonoscope'}]", "start": 1,"rows": 2500,"sort": "productid","sortorder": "asc"}

Notes:

- The API Endpoint returns all data when no filters are defined up to the row limits noted above.
- PRODUCTID column is the unique key for each record returned.
- PRODUCTID and RID are the default columns returned when no value is specified in the "displaycolumns" parameter.
- Filtering on date range using only the "from" date parameter considers today's date as the "to" date unless specified.
- Filtering on date range using only the "to" date parameter considers the earliest date in the database as the "from" date unless specified.
- The following are code examples for using the POST operation in various languages:

- Python
- PHP
- Java
- C#

Response Class

Model | Example Value

```
{
  "MESSAGE": "string",
  "RESULTCOUNT": 0,
  "APIRETURNCODE": 0,
  "RESULT": {
    "COLUMNS": [
      "string"
    ],
    "DATA": [
      [
        "string"
      ]
    ]
  }
}
```

Response Content Type application/json

Parameters

Parameter	Value	Description	Parameter Type	Data Type
payLoad	**(required)**	**payload input**	formData	string

Try it out!

Response Messages

API Return Code	Reason
400	Success
401	Authorization has been denied for this request
402	The input provided is not a valid JSON format
403	The input provided is not in the expected payload structure
406	The payload displaycolumns does not match with the datagroup
407	The payload sort does not match with the datagroup
408	The payload start should be numeric value and should be greater than 0
409	The payload rows should be numeric value and should be greater than 0 or blank if you intend to retrieve all rows
411	The payload sortorder is invalid
412	No results found

API Return Code	Reason
413	The payload filters are invalid
414	The payload is missing
417	The payload rows should be less than or equal to 2500 if you intend to retrieve Code Information data
418	Too many rows requested at this time. Please fetch rows fewer than 5000

[BASE URL: /rest/iresapi/ , API VERSION: 2.0.0]